

Hot Runner Balance / Temperature DOE

Plastic Injection Molding Technical Reference

For engineers, processors, setup technicians, tooling engineers, quality engineers, and manufacturing leaders

Hot runner balance / temperature DOE: controlled thermal changes, fixed machine process, cavity-by-cavity response

Purpose: distinguish thermal tuning opportunity from geometry, valve timing, heater/TC failure, shear imbalance, or process instability.

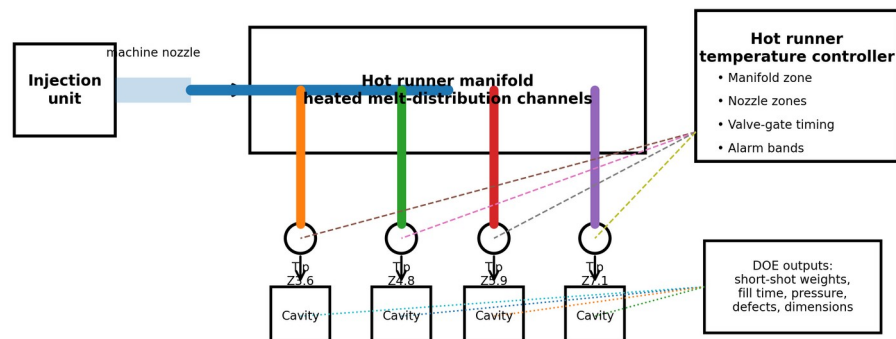


Figure 1 - DOE architecture: controlled thermal factors, fixed machine/process baseline, and cavity-by-cavity responses.

Document status: technical draft prepared from public manufacturer guidance, published technical papers, standards references, and practitioner discussions. It is not an independently peer-reviewed standard operating procedure. Before release to production, validate this document against the resin supplier datasheet, hot runner OEM manual, machine OEM safety documentation, internal quality system, and local regulatory requirements.

Executive Summary

A Hot runner balance/temp DOE is a controlled, cavity-by-cavity experiment used to determine whether nozzle, manifold, branch, and valve-gate thermal settings are contributing to fill imbalance in a hot-runner mold, and whether controlled temperature profiling can reduce that imbalance without creating a weaker, narrower, or unsafe process window. The core output is not simply a new set of hot-runner setpoints. The real output is a quantified understanding of each cavity's response to hot-runner temperature, separated from machine instability, non-return-valve variation, material condition, mold cooling, venting, gate steel, and valve timing effects.

The difference between hot runner and cold runner tooling is fundamental. In a cold runner system, the sprue and runners freeze every shot, the effective runner diameter changes during fill, and the runner is ejected or reground. In a hot runner system, the melt-distribution system remains molten, contains heaters and thermocouples, may include valve pins, and can be adjusted zone-by-zone. That adjustability is powerful, but it also creates a failure mode: a process can appear balanced only because individual cavities are being fed different melt temperatures. This may solve a fill-weight problem while creating different shrinkage, gloss, crystallinity, molecular weight, color, residence history, or weld-line strength between cavities.

The recommended business rule is: use a hot runner balance/temp DOE to identify and quantify balance behavior, not to hide tooling defects. Small validated temperature offsets can be acceptable. Large offsets, unstable responses, or offsets that violate the material processing window should trigger a tooling, hot-runner, valve-gate, material, or machine investigation.

1. Definition and Overview

1.1 Precise technical definition

A Hot runner balance/temp DOE is a statistically structured injection-molding experiment in which selected hot-runner temperature factors - typically manifold zones, branch zones, nozzle/tip zones, and sometimes valve-gate timing-related thermal states - are varied within validated limits while the machine process, material condition, mold cooling, and shot-size strategy are held constant. The response is measured by cavity-identifiable outputs such as fill-only short-shot weight, fill percentage, fill time, peak injection pressure, cavity pressure traces, full-cycle part weight, dimension, warp, flash, short-shot rate, gate appearance, gloss, weld-line quality, and process capability.

The DOE is usually performed after prerequisite scientific molding studies have established stable melt temperature, a stable shot size and cushion, a stable in-press viscosity curve, gate seal, pressure loss, hold-pressure response, and non-return-valve repeatability. Without those prerequisites, hot-runner temperature may be blamed for variation that actually comes from the barrel, screw, check ring, dryer, cooling circuit, or valve-gate mechanics.

1.2 What successful outcomes look like

A valid DOE produces three deliverables: a baseline balance map, a temperature-response model, and a confirmation run. The baseline map shows which cavities fill heavy or light at a flat temperature profile. The response model shows how each cavity or flow group responds when specific zones are changed. The confirmation run proves that the chosen profile improves balance during fill-only and during the full packed cycle.

The strongest evidence comes from a combination of: repeatable short-shot weights, repeatable cavity pressure curves, visual short-shot fill-front progression, stable actual-zone temperatures, stable cushion, stable recovery, and full-cycle part data. A short-shot-only win is not sufficient if the packed parts show unacceptable gate blush, flash, warpage, dimensional spread, crystallinity variation, degradation, or weld-line weakness.

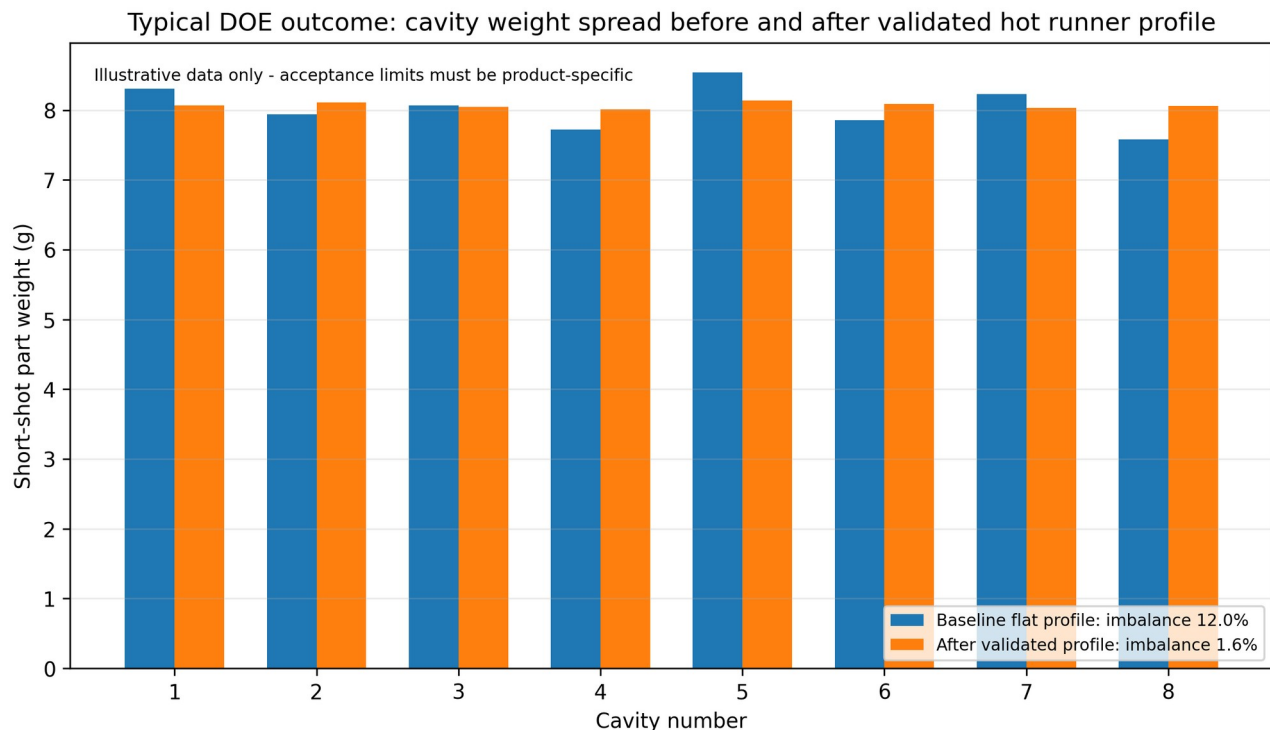


Figure 2 - Illustrative cavity weight outcome. A real study must use actual cavity IDs, calibrated scales, and validated acceptance limits.

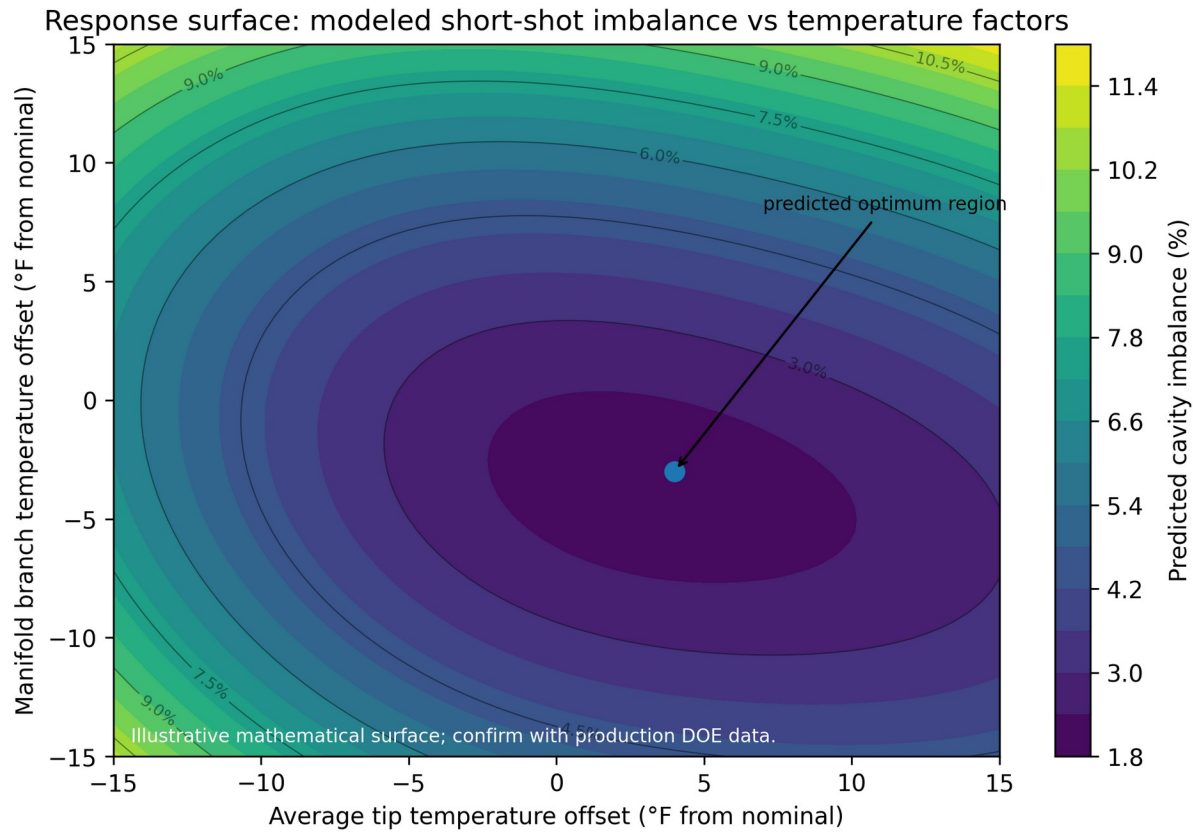


Figure 3 - Example response-surface concept. The minimum imbalance zone is not automatically the best process if it requires unsafe temperature offsets or causes quality defects.

1.3 Core response variables and calculations

Response	Definition	How to use it	Risk if misused
Short-shot weight by cavity	Weight of each cavity at an intentional fill-only condition, typically 60-95% full with no pack/hold.	Primary balance metric because it captures relative flow during fill.	If transfer is inconsistent or the shot is too small, results can reflect machine instability instead of runner balance.
Fill imbalance %	(Maximum cavity short-shot weight - minimum cavity short-shot weight) / average cavity short-shot weight x 100. Some companies use max deviation from average; define it before the study.	Single-number benchmark for baseline and confirmation.	No universal acceptance limit exists; product risk, cavitation, material, and tolerance define the target.
Cavity pressure overlay	Pressure trace at the end of fill, peak pack, gate seal, and cooling.	Separates fill balance from pack/hold imbalance and valve-gate timing issues.	Pressure sensors must be zeroed, maintained, and located appropriately.
Full-cycle part weight	Weight after pack/hold/cooling at production settings.	Validates the production result after the fill-only balance work.	A balanced full weight can hide imbalanced fill if hold pressure backfills late cavities.
Dimensional spread	Cavity-to-cavity dimensions, roundness, concentricity, flatness, thread fit, or sealing feature data.	Final quality confirmation.	Dimensional data alone does not identify whether the root cause is fill, pack, cooling, steel, or ejection.
Actual zone temperature	Controller PV, independent thermocouple reading if accessible, and alarm history.	Confirms the DOE factor was actually delivered.	Setpoint is not proof; bad thermocouple location or heater output can make setpoint misleading.

2. Root Causes: Why a Hot Runner Balance/Temp DOE Is Done

A hot runner balance/temp DOE is run when multi-cavity or multi-gate fill behavior is not acceptably repeatable or not acceptably uniform. Symptoms include cavity-to-cavity short shots, some cavities flashing while others are short, inconsistent gate vestige, cavity-specific burn marks, part-weight spread, dimensional capability loss, different gloss by cavity, weld-line movement, core shift, sink/void variation, or a process that requires excessive pack pressure to cover late-filling cavities. Published hot-runner and mold-balance references consistently treat system balance as a mold-qualification requirement, but also recognize that acceptable balance and measurement procedure are application-specific, not a universal number [R1].

2.1 Material factors

Most thermoplastic melts are non-Newtonian and shear-thinning. As shear rate rises, apparent viscosity drops; as temperature rises, viscosity generally drops further. Hot runner balance is therefore a rheology problem, not just a steel-layout problem. If one cavity receives a hotter or more highly sheared stream, it may fill faster and pack differently than a cavity receiving colder or lower-shear melt. This is why a geometrically balanced runner can still fill unbalanced.

Material lot, colorant, regrind, moisture, fillers, flame retardants, lubricants, nucleating agents, impact modifiers, and recycled content can all change viscosity, thermal stability, pressure loss, shear heating, and freeze-off behavior. Capillary rheology under ASTM D3835 or comparable methods can quantify viscosity versus shear rate, temperature sensitivity, and dwell-time stability, which is directly relevant when a thermal DOE is close to the edge of a resin's processing window [R23].

Polymer / material class	Expected influence on hot runner temp DOE	Special control actions
PP and PE	Generally broad processing window and strong shear-thinning. Semi-crystalline shrinkage makes cavity-to-cavity thermal history important. Nucleated or filled grades can react differently than neat grades.	Verify shrink/warp and dimensions after fill balance. Do not assume balanced short shots equal balanced shrinkage.
ABS	Amorphous. Surface gloss, weld-line appearance, splay, and color can move with melt temperature and shear. Moderate thermal sensitivity depending on grade and color package.	Use visual acceptance boards by cavity. Watch gloss and gate blush when offsetting tips.
PC	Higher viscosity and more temperature-sensitive than many commodity resins. Moisture and residence time can drive splay, brittleness, yellowing, or molecular degradation.	Dry correctly, minimize residence time, and avoid using high hot-runner temperatures to compensate for a restriction.
PA / nylon	Moisture and drying state strongly affect viscosity and properties. Semi-crystalline freezing behavior can narrow the practical hot-runner window. Long residence at high temperature can generate decomposition products.	Use resin supplier drying and hold-up-time limits. Confirm mechanical properties for critical parts.
POM / acetal	Thermally sensitive; degradation can be hazardous. Hot spots and long hold-up are high risk.	Follow resin/OEM safety rules; avoid exploratory overheating.
PVC	Heat-sensitive; shear and temperature must be controlled to avoid degradation. Published studies show filling balance can be optimized through injection rate and simulation while keeping temperatures below degradation limits [R17].	Use conservative temperature limits and ventilation. Do not use broad thermal offsets as first-line correction.
Glass-filled materials	Higher viscosity, fiber orientation, abrasion, and potential gate wear. Balance changes may reflect worn gates or preferential fiber alignment, not only temperature.	Inspect gates/tips for erosion. Confirm dimensions, warp, and mechanical orientation-sensitive properties.
LCP and highly filled compounds	Very shear-sensitive and orientation-sensitive; small flow path differences can produce large property differences.	Involve hot-runner supplier early. Avoid dead spots and residence-time traps.

2.2 Process parameters

Injection speed changes shear rate, viscous heating, fill time, pressure loss, and the distribution of hotter and colder melt layers through each runner split. A DOE run at one speed is not automatically valid at another. If the in-press viscosity curve shows a major pressure change between speeds, the hot-runner balance profile may also be speed-specific. Injection pressure must be sufficiently above actual pressure demand; otherwise pressure-limited filling can falsely appear as a hot runner balance issue.

Mold temperature and cooling rate affect frozen-layer growth, gate freeze, gloss, crystallinity, shrinkage, and warp. They have less influence on fill-only short-shot weight than they do on packed-part dimensions and appearance, but they still matter when the short-shot study is not truly fill-only or when gates freeze early. Mold temperature control errors are a major reject driver, and modern systems monitor and regulate water flow, differential temperature, and circuit behavior to keep process conditions repeatable [R12].

Parameter	How it can distort or drive balance results	DOE control requirement
Injection speed / fill time	Changes shear rate and melt heating. A faster speed may make a high-shear path fill first; a slower speed may expose cold restrictions.	Lock fill time during the DOE. Record actual fill time and peak pressure every shot.
Pressure limit	A low limit masks flow differences because all cavities become pressure-starved.	Set pressure limit high enough for fill-only trials while remaining safe for tool and machine.
Transfer position	If transfer occurs before a repeatable fill-only endpoint, cavity weights become non-comparable.	Use a fixed fill-only shot size or fixed transfer position, no pack/hold.
Cooling time	Can affect part removal, gate freeze, crystallinity, and packed dimensions.	Keep fixed during full-cycle confirmation; short-shot fill-only parts are normally ejected before full pack analysis.
Back pressure and screw RPM	Change melt homogeneity, melt temperature, recovery time, and residence history.	Lock recovery profile after melt-temperature verification.
Decompression / suckback	Can pull melt from open tips, create drool/stringing changes, and change first-injection response.	Lock and document. Avoid changing during DOE unless it is a formal factor.

2.3 Mold design factors: hot runner vs cold runner

Cold runner mold: sprue + solidifying runner

Hot runner mold: heated manifold + drops/tips

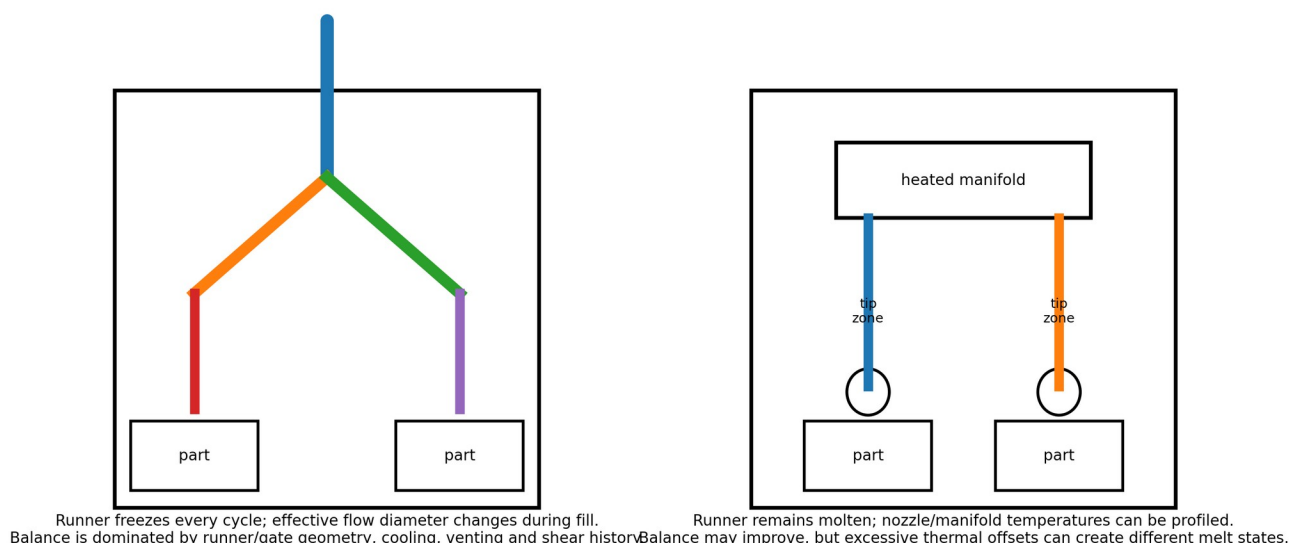


Figure 4 - Cold runner versus hot runner behavior. Thermal adjustability is unique to the hot runner, but it must not be used to hide mechanical imbalance.

Cold runner balance is mostly controlled through sprue, runner, gate geometry, steel temperature, venting, and part geometry. The runner cools and solidifies each cycle, so the effective runner diameter can decrease during fill. That increases shear and can amplify shear-induced temperature distribution through runner splits.

Hot runner balance adds manifold channel geometry, heater layout, thermocouple location, insulation, expansion, nozzle type, tip style, valve pin movement, controller performance, leakage paths, dead spots, gate cooling, and residence time. A hot runner can be naturally balanced by equal flow length and channel diameter from nozzle to each cavity, but natural geometric balance alone does not guarantee rheological balance. Mold-Masters describes natural balance as equal flow length and diameter from machine nozzle to each cavity, while also noting that shear and geometry effects still matter [R2]. Beaumont similarly emphasizes that hot runners can exhibit shear-induced imbalances and also add steel-related variables such as temperature fluctuation, gate-size variation, plugs/mismatch in channels, thermal expansion, and heater/thermocouple functionality [R4].

Mold feature	Cold runner effect	Hot runner effect	DOE implication
Sprue/runner	Solidifies every shot. Runner weight and freeze profile affect flow and regrind.	Replaced by heated manifold channels and drops that stay molten.	Cold runner balance cannot be tuned zone-by-zone; hot runner can, but setpoint changes affect melt state.
Gate size/location	Gate freeze and pressure loss dominate packing and fill balance.	Gate tip temperature, gate insert cooling, valve-pin shutoff, and gate wear add variables.	If one gate is worn or plugged, temperature DOE will only mask the real root cause.
Wall thickness variation	Thicker regions may fill/pull flow differently and pack differently.	Same, plus hot runner can feed regions at different thermal histories.	Use short-shot progression photos to distinguish runner imbalance from part-flow imbalance.
Venting / air traps	Late cavities may be gas-blocked and look like poor balance.	Same, especially high cavitation tools with small vents and high fill speeds.	Clean and verify vents before DOE. Burns or inconsistent short shots are not a temperature-only problem.
Cooling circuits	Affects freeze-off, shrink, warp, and dimensions.	Same; also affects hot tip thermal boundary and gate freeze.	Document water temperature, flow, and delta-T. Do not change during DOE.
Valve gates	Not present except special cold-runner valves.	Pin open/close timing, stroke, wear, hydraulic/pneumatic/electric actuation and sequence can dominate fill order.	Separate valve timing studies from pure temperature DOE unless the experiment is explicitly multi-factor.

2.4 Machine factors

Machine-side variation can create false hot-runner conclusions. A leaking non-return valve or worn screw can produce shot-to-shot fill variation. Barrel heater errors can shift melt temperature. Excessive residence time can degrade resin. Pressure control or velocity-control errors can change fill. Clamp issues can affect venting, flash, or parting-line seal, which can then alter cavity fill. Before a DOE, the press must be treated as a controlled measurement system.

Machine factor	Failure mode	Pre-DOE control
Screw/check ring/NRV	Cushion drift, inconsistent transfer pressure, inconsistent short-shot weights.	Run NRV repeatability and cushion study. Reject DOE if shot-to-shot variation is not stable.
Screw design/wear	Poor melting, unmelted pellets, fiber breakage, melt-temperature variation.	Verify melt temperature, recovery time, and screw condition for chronic issues.
Back pressure/screw RPM	Changes melt homogeneity and shear heating.	Lock after melt verification; document recovery profile.
Clamp tonnage/platen parallelism	Flash in some cavities, blocked vents, dimensional spread.	Verify clamp is adequate and mold seating is clean; do not use clamp to compensate for filling imbalance.
Machine size	A small shot in an oversized barrel increases residence time and degradation risk.	Confirm shot size is appropriate for barrel capacity and resin hold-up guidance.

2.5 Auxiliary equipment roles

Auxiliary equipment	Role in DOE	Failure mode / false conclusion
Dryer / hopper / loader	Controls moisture and material feed consistency.	Moisture shifts viscosity, causes splay, brittleness, degradation, and weight variation.
Chiller / thermolator / TCU	Controls mold water temperature, flow, and heat extraction.	Circuit imbalance changes gate freeze and shrink; apparent hot-runner balance may actually be cooling imbalance.
Hot runner controller	Delivers the experimental factors and records setpoint/PV/output/alarm state.	Bad thermocouple, swapped zones, open heater, poor tuning, slave mode, or wrong zone map invalidates DOE.
Valve-gate sequencer	Controls gate opening and closing when valve gates are present.	Timing or stroke variation can dominate cavity filling and make temperature profiling

Robot / sprue picker / conveyor	Must not distort short shots or change cycle time during data collection.	misleading. Part damage, cavity mix-up, or cycle-time variation ruins traceability.
Scale / data acquisition	Provides the measurement system.	Insufficient resolution or no cavity ID destroys statistical confidence.

3. Real-World Examples and Case Studies

This section uses public case studies, manufacturer examples, technical papers, and forum discussions. No private interviews were conducted for this document. Forum posts should be treated as practitioner anecdotes, not peer-reviewed evidence; they are useful because they show how technicians and engineers actually reason through balance problems on the floor.

Source / setting	Observed issue	Method or lesson	Practical takeaway
RJG / ANTEC valve-gate sequencing [R5]	Hot runner molds can have consistent cavity imbalance and shot-to-shot variation. Molders often attempt to thermally balance by adjusting zone temperatures until cavities fill simultaneously.	The paper warns that thermal balancing can create dissimilar melt temperatures between cavities and presents valve gates with cavity pressure sensing for balance and pressure control.	Temperature profiling is not always the best control strategy. For valve-gated tools, valve sequencing and cavity-pressure control may be more robust.
Beaumont mold balance work [R4]	Studies across multi-cavity molds separated steel imbalance from shear-induced imbalance. The article notes hot runners also add variables such as heater/TC placement and temperature fluctuation.	Short-shot and flow-group analysis were used to diagnose whether imbalance was caused by shear or steel-related defects.	Do not assume a hot runner issue is purely thermal. Map flow groups and isolate shear versus steel effects.
Beaumont / INCOE Opti-Flo example [R3]	Shear-induced imbalances affected mold efficiency and product quality; some conventional manifolds required added cycle time to meet dimensions.	Melt-management technology was used to manage shear distribution in hot runner systems.	If balance requires major thermal offsets, geometry or melt-management changes may be better than process compensation.
Moldex3D HRS analysis [R6]	High-cavitation hot runner designs are expensive to iterate physically.	Hot Runner Steady analysis predicts flow rate, balance ratio, and non-uniform temperature distribution after accumulated shear heating.	Use simulation before steel changes and before expensive trial-and-error when the runner layout is complex.
Moldex3D Felli case [R7]	Flow imbalance and core shift in a single-cavity hot-runner application.	FSI/core-shift simulation, valve-gate hot-runner strategy, product-thickness changes, and cavity/core plate temperature changes were used.	Hot runner balance can be coupled to part stiffness, core deflection, and thermal gradients - not just cavity weight.
Eng-Tips technician discussion [R19]	Repeated pattern: some cavities full, one partly full, one very short.	Responder pointed to cushion, gate/runner size, venting, temperatures, shot size, injection speed, or restrictive flow path.	Stable imbalance pattern needs mechanical/process checks before temperature-only corrections.
CR4 troubleshooting discussion [R20]	Intermittent short shots and flash symptoms.	Responders asked whether the same cavity repeats, whether runner balance had been checked, and cautioned against blind pressure/speed tweaking.	Ask whether the symptom is stable by cavity, shot-to-shot random, or linked to machine recovery.

4. Diagnostic Techniques

4.1 How to know whether a hot runner balance/temp DOE was done after production is running

After production has already started, evidence of a completed DOE should be visible in documentation and in the process-control trail. Visual inspection alone cannot prove that a DOE was done. The strongest evidence is a complete DOE package: cavity-numbered short-shot samples, cavity-weight spreadsheet, zone setpoint history, actual PV history, process sheet revision, confirmation-run data, and a reaction plan defining what to do if cavity balance drifts.

Evidence type	What to look for	Interpretation
Setup sheet / process sheet	Non-flat hot runner temperature profile with revision date and reason.	Suggests a DOE or trial occurred; does not prove it was controlled.
Controller history	PV and output traces for each zone, alarm log, autotune record, zone map.	Confirms actual delivered thermal profile and catches bad zones.
Short-shot sample set	Cavity IDs molded, no pack/hold, fill-only progression photos, weights.	Best physical evidence of balance evaluation.
Quality records	Cavity-by-cavity part weights, dimensions, visual defects, capability data.	Shows whether the DOE result carried into production.
Tool maintenance notes	Gate clean/repair, thermocouple repair, heater replacement, valve-pin service.	May explain why a profile changed or why DOE results no longer apply.
Operator notes	Cycle interruptions, material changes, dryer alarms, water-flow adjustments.	Useful for identifying confounders and false conclusions.

4.2 Visual, microscopic, and metrology diagnostics

Symptom	Possible balance/temp meaning	Verification method
Same cavity repeatedly short	Low local melt temperature, restricted tip, plugged gate, bad valve opening, blocked vent, low-flow branch.	Short-shot series, tip/gate inspection, thermal verification, valve stroke check.
Some cavities flash while others short	Unbalanced fill/pack, clamp or venting issue, early gates overpacked while late gates underfilled.	Fill-only study plus full-cycle cavity pressure and parting-line inspection.
Gate blush / halo / stringing	Tip too hot, excessive shear, wrong gate design, insufficient cooling at gate.	Vary tip temperature cautiously, inspect gate land, compare controller PV to independent check.
Burns at flow end	Air trap/venting limitation, too high speed, degraded melt, excessive shear heat.	Vent inspection, short-shot progression, decompression and drying checks.
Different gloss by cavity	Different melt or mold temperature, different packing, different flow orientation.	Gloss measurement, mold surface temp mapping, cavity pressure overlay.
Weld-line movement or weakness	Fill-front timing changed by hot runner profile or valve sequence.	Short-shot photos, tensile/impact test at weld line, pressure and fill-time overlay.
Splay or silvering by cavity	Moisture, degradation, air entrainment, local overheating, decompression/drool.	Dryer dew point, material moisture, residence-time calculation, tip inspection.

Microscopy can help evaluate weld-line structure, fiber orientation, contamination, burn/degradation, delamination, or void morphology. Polarized light can reveal stress patterns in transparent materials. Microtome sections can reveal knit-line defects or glass-fiber distribution. These methods are confirmatory; they do not replace a controlled fill-only balance study.

4.3 Simulation and predictive methods

Moldflow, Moldex3D, and similar CAE tools can predict pressure loss, shear rate, temperature distribution, gate contribution, fill time, weld lines, air traps, and hot-runner flow rates. Autodesk Moldflow Runner Balance analysis is intended to determine optimum runner section volume for balanced filling [R23 is rheology; see Autodesk source in reference notes if available]. Moldex3D Hot Runner Steady analysis specifically targets hot-runner layouts and can predict flow rate and balance ratio without always simulating every cavity in full detail; it also accounts for accumulated cycle-by-cycle viscous heating to establish more realistic initial runner temperature distribution [R6].

Simulation is not a substitute for press validation because real tools contain thermocouple location errors, manifold machining tolerances, valve-pin wear, heater aging, leaks, cooling variation, resin lot shifts, and machine-specific dynamics. The best practice is closed-loop: simulate the hot-runner layout, run a baseline DOE, tune or repair, and feed the measured results back into the model.

4.4 Alternatives that can equal or outperform a temperature DOE

Alternative method	When it is better than temperature profiling	Why
Cavity pressure control / dynamic feed control	High-value medical, optical, automotive, or tight-tolerance parts with sensors and valve gates.	Controls what happens in the cavity rather than inferring from external temperature.
Valve-gate sequencing DOE	Sequential-fill tools, family tools, long-flow parts, or multi-gate single cavities.	Timing can correct fill-front control without giving cavities different melt temperatures.
Mechanical gate/nozzle adjustment	When one or two cavities are consistently high/low and inspection shows gate or tip variation.	Corrects root cause instead of process compensation.

Melt management / runner redesign	When shear-induced imbalance is large or repeatable across flow groups.	Redistributes shear history or changes geometry to make balance robust.
Moldflow/Moldex3D design iteration	Before cutting steel or when cavitation is high.	Reduces trial cost and screens design options.
In-mold thermal/pressure instrumentation	When normal press data cannot separate fill, pack, and cooling effects.	Provides direct process observability.

4.5 Non-destructive testing options

There is no universal non-destructive test that directly grades internal flow-line severity for every polymer and part geometry. The available methods evaluate consequences of flow imbalance: voids, density variation, orientation, stress, delamination, weld-line weakness, or internal defects. Use NDT only after correlating it to destructive tests and production failures.

Method	Useful for	Limitations
X-ray / CT scanning	Voids, sinks, internal short shots, inserts, core shift, density-related defects.	Expensive; may not reveal molecular orientation or weld strength directly.
Ultrasonic / acoustic inspection	Voids, delamination, some weld-related discontinuities.	Requires geometry-specific calibration and coupling.
Infrared thermography	Cooling non-uniformity, hot spots, short-shot thermal patterns soon after ejection.	Surface-dominant; timing-sensitive.
Polarized light	Residual stress in transparent amorphous parts.	Limited to transparent/translucent parts and qualitative unless calibrated.
Optical microscopy	Surface flow lines, burn marks, weld line morphology, fiber orientation on sections.	Usually requires sectioning if internal features are evaluated.

5. Preventive Measures Before and After Running the DOE

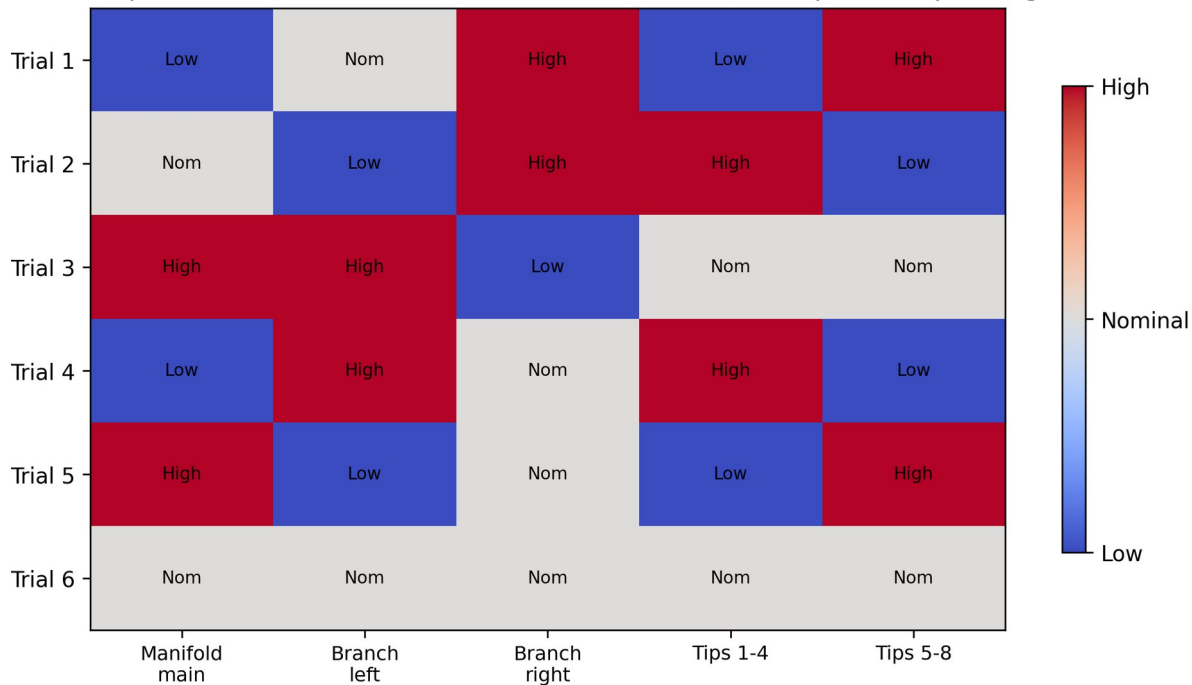
5.1 Material selection and preparation

- Choose a resin grade with a practical viscosity window for the part wall thickness, flow length, cavitation, and gate design. Do not select only by nominal melt flow rate; MFR is low-shear and may not represent injection shear conditions.
- Use capillary rheology, resin supplier processing data, and previous in-press viscosity curves for critical programs.
- Control drying, moisture, regrind ratio, colorant percentage, and material lot. These factors can move the response surface and invalidate a profile.
- Use lubricants or flow promoters only with engineering approval. They may improve fill but can damage weld strength, paintability, seal performance, medical compliance, or long-term properties.
- For hot runners processing heat-sensitive materials, compute residence time in barrel plus hot runner and follow resin supplier hold-up limits. Long dwell at high temperature can generate decomposition products [R10].

5.2 Process optimization: what the DOE tells you

A good DOE tells you whether cavity imbalance is temperature-responsive, which zones influence which cavities, how steep the response is, whether the response is linear or interacting, and whether the effect is stable over time. Use that information to set the smallest temperature offsets that meet quality requirements. If the model says a 35 F tip offset is required for a basic balance target, the correct answer is usually not to run a 35 F offset forever; the correct answer is to inspect the steel, gate, tip, valve pin, heater, thermocouple, cooling, and flow path.

Example blocked/fractional DOE matrix for hot runner temperature profiling



Use actual level sizes from resin and hot-runner OEM limits; randomize only when safe and practical.

Figure 5 - Example DOE matrix. Use resin/OEM limits and plant quality rules to define levels.

DOE result	Interpretation	Action
Strong cavity response to one tip zone	Local gate/tip thermal sensitivity or local restriction.	Confirm actual tip behavior; use small offset if safe; inspect gate/tip if required offset is large.
All cavities shift together with manifold temp	Global viscosity/melt-temperature effect.	Adjust global melt or manifold only after verifying barrel melt temp and residence time.
Opposite cavities respond to branch zones	Branch manifold influence or flow-group behavior.	Use branch profile if stable; investigate machining and insulation if effect is excessive.
Response changes over time	Heat soak, degradation, valve timing drift, heater control problem, material condition shift.	Do not lock profile. Troubleshoot stability.
No temperature response but imbalance persists	Geometry, valve, venting, gate, process, or machine issue.	Escalate to root-cause troubleshooting.

5.3 Cold runner vs hot runner parameter adjustments

Adjustment	Cold runner mold	Hot runner mold
Melt temperature	Global barrel/nozzle effect. Can reduce viscosity but increases cycle, shrink, degradation risk.	Global effect plus interaction with manifold/tips. Must coordinate barrel melt temp and runner setpoints.
Runner temperature	Not directly adjustable; runner cools and freezes each cycle.	Directly adjustable by zone. This is the DOE factor but also a possible source of cavity-to-cavity melt differences.
Injection speed	Changes shear heating in sprue/runner and cavity; may improve balance or burns.	Changes shear heating in barrel, manifold, drops, and gates. Speed profile must be locked for temperature DOE.
Gate size	Steel change or insert repair; major driver of balance and pack.	Gate/tip insert condition plus local hot tip temperature. Gate wear can be masked by temperature.
Pack/hold	Can compensate partially after fill but may overpack early cavities.	Same, plus valve-gate close timing may isolate cavities.
Cooling	Changes runner freeze and gate seal.	Changes gate/tip thermal boundary and part shrink; hot runner remains molten.

5.4 Mold design improvements

- Use natural balance where feasible: equal flow length, equal channel diameter, equal branch count, and symmetrical thermal boundary conditions.
- Avoid sharp turns, dead spots, abrupt diameter transitions, and material hang-up regions. Curved and polished runner paths reduce stagnation and shear concentrations.
- Select gate location to support balanced fill-front progression, adequate venting, and reasonable weld-line position.
- Maintain uniform wall thickness where possible. Where thickness changes are unavoidable, use flow leaders, ribs, or gate strategy intentionally rather than letting the runner compensate randomly.
- Vent end-of-fill regions, weld-line regions, and runner dead ends. Blocked vents can look exactly like a temperature balance issue.
- Design cooling around gates and tips deliberately. Hot tips surrounded by cold steel, poor insulation, or uneven water circuits create thermal sensitivity.

5.5 Machine calibration and maintenance

- Calibrate barrel heats, nozzle heat, pressure transducers, screw position, and injection velocity before qualification work.
- Run non-return-valve repeatability. A hot-runner DOE is invalid if the injected volume is not repeatable.
- Inspect screw/barrel wear when there is chronic melt-temperature spread, black specks, unmelt, inconsistent recovery, or process drift.
- Maintain press hydraulic/electric response. Velocity-to-pressure transfer and pressure limiting must be repeatable.
- Verify clamp tonnage, platen parallelism, mold mounting, and parting-line condition if flash and short shots occur together.

5.6 In-process adjustments and reaction plan

A production reaction plan must define what technicians may change, how much they may change it, and what evidence is required before a change is made. Without a reaction plan, a validated DOE profile becomes tribal knowledge and eventually degrades into random knob turning.

Production event	Allowed response	Escalation trigger
One cavity trends light over 30 minutes	Check cavity ID, material feed, dryer, water flow, gate drool, and controller PV before changing temperature.	Escalate if the same cavity remains out after basic checks or if zone output is abnormal.
Hot runner alarm	Stop or quarantine according to quality plan. Do not continue running critical parts on an alarmed zone.	Maintenance/OEM if heater or TC fault persists.
Material lot/color change	Run reduced confirmation: melt temp, short-shot balance, visual, weight.	Full DOE if response shifts beyond documented window.
Startup after long stop	Purge and re-stabilize; run documented startup scrap count and cavity check.	Escalate if degraded material or black specks persist.
Unplanned temperature change	Return to validated setpoint; segregate product made during deviation.	Quality disposition and process engineering review.

5.7 Post-processing solutions: cold runner and hot runner tools

Post-processing should be treated as containment, not root-cause correction. Polishing, painting, pad printing, trimming, annealing, or secondary machining can reduce customer-visible symptoms, but they do not repair fill imbalance. For cold-runner tools, post-processing may hide runner/gate vestige or cosmetic defects caused by freeze-off and poor venting. For hot-runner tools, it may hide gate blush, stringing marks, color streaks, or gloss variation caused by tip temperature or residence-time problems. Use secondary operations only after a documented engineering decision confirms that mechanical properties, regulatory requirements, and cost targets are still acceptable.

5.8 Redesign strategies

- Modify part geometry to remove flow restrictions, sharp thickness transitions, unventable pockets, and long isolated flow lengths.

- Change gate location, add or remove gates, or switch from open tip to valve gate when fill-front control is the dominant issue.
- Change runner diameter, branch layout, manifold level changes, or melt-management features when shear-induced flow groups dominate.
- Use replaceable gate inserts or tip inserts on abrasive materials to preserve balance over tool life.
- Add cavity pressure sensors on high-value programs where pack control and gate-seal timing are critical.
- Involve the hot runner OEM, mold maker, resin supplier, process engineer, and quality engineer before steel changes. A balance problem is cross-functional by nature.

6. Recommended DOE Procedure

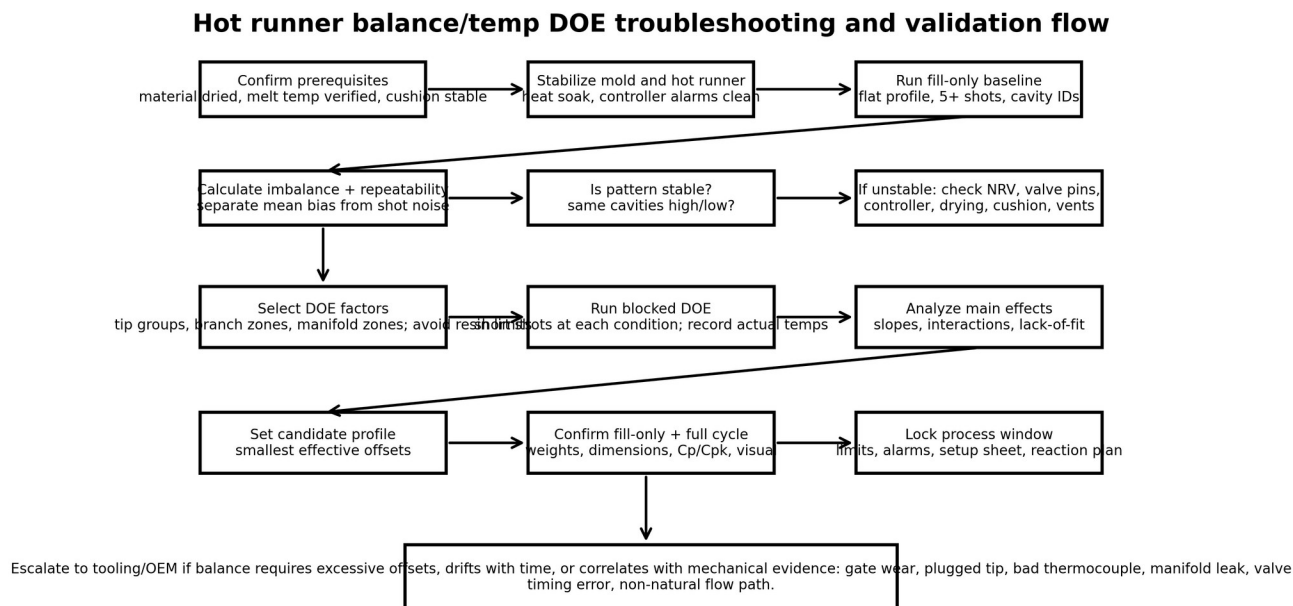


Figure 6 - Recommended workflow from prerequisites through confirmation and escalation.

6.1 Safety and setup gate

- Follow lockout/tagout and machine guarding requirements. Injection molding machines are covered by safety standards such as ISO 20430 for machine safety requirements [R14].
- Hot runner systems contain high-temperature surfaces, pressurized molten polymer, electrical heaters, thermocouples, hydraulic/pneumatic/electric valve actuators, and possible decomposition products. Use OEM manuals and plant PPE rules.
- Do not run the DOE with bypassed guards, active hot-runner alarms, unknown zone mapping, unknown material, or unverified cooling circuits.
- When processing PVC, POM/acetal, flame-retarded compounds, or other heat-sensitive materials, define maximum temperature, maximum dwell, purge procedure, and ventilation before starting.

6.2 Prerequisites before running the DOE

Prerequisite	Acceptance check	Why it matters
Material condition	Correct resin, lot, colorant, drying time, dew point, moisture where applicable.	Material state changes viscosity and degradation risk.
Melt temperature verification	Air shot or validated probe method consistent with safety plan.	Setpoint is not melt temperature.
Shot size/cushion study	Cushion stable; shot uses appropriate barrel capacity.	Injection volume must be repeatable.
NRV repeatability	Acceptable shot-to-shot weight/position	Leaking check ring can mimic balance

	repeatability.	variation.
In-press viscosity curve	Stable curve; selected fill speed in robust region.	Balance response is speed/shear dependent.
Pressure loss study	Machine, nozzle, sprue bushing, runner, gate, and cavity pressure demand understood.	A pressure-limited process invalidates balance conclusions.
Gate seal / hold-pressure curve	Known gate seal and packing response.	Full-cycle confirmation must not overpack early cavities.
Cooling verification	Water temperature, flow, delta-T, and circuit hookup documented.	Cooling variation changes gate freeze and dimensions.
Hot runner health	Zone map verified; no alarms; PV/setpoint/output plausible; valve pins tested.	DOE factors must be real and repeatable.

6.3 Baseline fill-only balance study

1. Install cavity identification method. Every part, short shot, runner slug, and sample bag must map to cavity number. Do not rely on memory.
2. Set all hot runner zones to the nominal resin/OEM profile unless a documented flat-profile baseline already exists.
3. Run until mold, hot runner, barrel, and cooling system are thermally stable. For high-mass manifolds, heat soak may require more time than normal startup.
4. Set pack/hold pressure and time to zero or to the lowest safe fill-only condition defined by the press/tool team. The goal is a controlled short shot, not a packed part.
5. Adjust shot size or transfer to create a 60-95% fill-only condition. Use the fill percentage that best separates cavities without damaging the tool or leaving fragile pieces trapped.
6. Collect at least five consecutive shots per condition after stabilization. More shots are required if shot-to-shot variation is high.
7. Weigh each cavity on a calibrated scale with sufficient resolution. Record actual fill time, peak pressure, cushion, recovery time, hot-runner PVs, water temperatures, dryer state, and visual notes.
8. Calculate imbalance, per-cavity mean, standard deviation, and coefficient of variation. Separate cavity bias from shot noise.

6.4 Temperature DOE design

The DOE should be as small as possible while still separating the major factors. In many production environments, a full factorial across every tip and manifold zone is impractical. Use engineering grouping: cavities that share a branch, tips that share a heater family, or flow groups that share shear history. A common structure is a blocked fractional factorial with low/nominal/high levels for selected zone groups, followed by a confirmation run at the predicted setting. Use actual PVs, not just setpoints, in the analysis.

Design choice	Use when	Notes
One-factor-at-a-time screening	Small tools, low risk, limited time.	Simple but misses interactions. Use only for preliminary diagnosis.
2-level fractional factorial	Multiple zone groups and limited press time.	Efficient for main effects; center points help detect curvature.
Response surface DOE	Strong nonlinear response or high-value tooling.	Better for optimizing setpoint window but requires more trials.
Sequential DOE	High-cavitation tools or heat-sensitive resins.	Start with safe screening, then refine only important factors.
Valve timing + temperature DOE	Valve-gated tools where timing and thermal behavior interact.	Must be planned carefully; do not mix casual timing changes into a pure temp DOE.

6.5 Analysis logic

- Plot cavity weight versus trial for each cavity. The plot should show whether the response is systematic or noise.
- Calculate main-effect slope for each factor: grams per degree or percent fill per degree. A high slope identifies sensitive zones or cavities.
- Check interactions. For example, a tip offset may only work when a branch manifold zone is also changed.
- Compare model residuals against shot sequence. Drift with shot order means the mold was still stabilizing or material was changing.
- Check actual zone output. A zone at 100% output or cycling abnormally may not be in control.

- Reject profiles that achieve balance only by exceeding resin limits, causing degradation, increasing scrap, or narrowing the process window.

6.6 Confirmation run

- Apply the candidate profile and return all non-temperature settings to the locked DOE baseline.
- Run fill-only confirmation: at least five stable short-shot cycles with cavity weights and visuals.
- Run full-cycle confirmation: production pack/hold/cool, part weights, critical dimensions, visual quality, gate appearance, and cavity pressure if available.
- Stress the window: run a small, documented process window check around the validated profile if quality rules allow. Confirm that acceptable parts are not produced only at a single fragile setpoint.
- Document the final profile with setpoint limits, actual PV expectations, alarm bands, recovery actions, and escalation criteria.

7. Industry Best Practices

There is no single ISO or ASTM procedure that defines a universal hot runner balance/temp DOE for production tooling. Industry practice draws from scientific molding, DOE methods, resin rheology standards, mold qualification discipline, in-mold pressure control research, hot-runner OEM guidance, and internal quality systems. Treat the DOE as a validated plant procedure with defined measurement system, sample size, data retention, acceptance rules, and reaction plan.

Organization / source type	Relevant guidance	How to apply
SPE / ANTEC / Polymer Engineering & Science	Published work on valve-gate sequencing and multi-cavity pressure control documents the benefit of in-cavity sensing and dynamic control [R5, R15].	Use cavity pressure and valve sequencing for high-value tools where thermal balance alone creates dissimilar melt conditions.
ASTM D3835 / ISO 11443-type rheology	Measures melt viscosity behavior across temperature, shear, and dwell-time conditions [R23].	Use rheology to set safe DOE levels and understand resin sensitivity.
ISO 294 series	Defines reproducible injection molding of thermoplastic test specimens [R24].	Use as a quality mindset: define and reproduce molding conditions rather than relying on undocumented adjustments.
ISO 20430	Machine safety requirements for injection molding machines [R14].	Keep DOE work inside guarded, documented machine safety boundaries.
Hot runner OEMs	Natural balance, temperature control, troubleshooting, and system maintenance guidance [R1, R2, R11].	Validate zone maps, thermocouple placement, heater health, valve operation, and resin compatibility.
Machine OEMs	Integrated mold temperature and hot-runner/cylinder heating controls improve repeatability and monitoring [R12, R13].	Capture data from integrated controls; trend deviations and alarm states.

7.1 Shop-floor rules from experienced molders

- Change one controlled variable at a time unless the DOE deliberately requires multi-factor trials.
- Never call a hot-runner temperature profile "validated" until it has passed both fill-only and full-cycle confirmation.
- Never tune a hot runner while the machine is pressure-limited, the cushion is drifting, or the material is not under control.
- Use cavity numbers, not visual guesses. A perfect-looking tray of parts can still have cavity-to-cavity imbalance hidden by pack pressure.
- If a cavity requires a large temperature offset, inspect hardware before accepting the setting.
- Record actual temperatures, not just setpoints. A bad thermocouple can make the setpoint fiction.
- Preserve the baseline flat profile data. It is the reference point for future troubleshooting after maintenance or resin changes.

8. Emerging Technologies and Research

Hot runner balancing is moving from static setpoint tuning toward instrumented, model-based control. The objective is not simply to reduce average imbalance, but to maintain balance automatically as resin lot, ambient temperature, water temperature, valve response, and machine dynamics drift.

Technology	Current / emerging value	Implementation caution
IoT-enabled hot runner controllers	Trend zone PV, setpoint, heater output, alarms, and energy over time. Detect heater aging or thermocouple drift before scrap.	Data must map correctly to physical zones. Bad zone naming destroys usefulness.
Cavity pressure and temperature sensors	Directly observe fill, pack, gate seal, and cooling in each cavity.	Adds tooling cost and maintenance; sensors must be protected and calibrated.
AI / machine learning optimization	Can model nonlinear temperature, pressure, speed, cooling, and material interactions from historical data.	Requires clean labeled data, cavity IDs, and guardrails. AI should not recommend unsafe temperatures or unapproved settings.
Digital twins / simulation-linked DOE	Use CAE to propose DOE factors and reduce press trials; update model with measured data.	Model accuracy depends on material data, boundary conditions, and realistic hot-runner thermal assumptions.
Valve-gate closed-loop control	Dynamic control of gate opening, closing, and pressure can balance flow without changing nozzle temperatures.	Needs robust sensors, validated control logic, and maintenance discipline.
Advanced manifold geometry / melt management	Curved channels, flow-path equalization, and shear redistribution can make balance more inherent.	Requires OEM/tooling involvement and may be difficult to retrofit.
Material science innovations	Nano-additives, self-healing polymer concepts, tailored rheology, and improved stabilizer packages may widen processing windows.	Additives can alter viscosity, shrink, color, weld strength, compliance, and recyclability; validate fully.

9. Troubleshooting Guide

Common sources of false determination in a hot runner balance/temp DOE

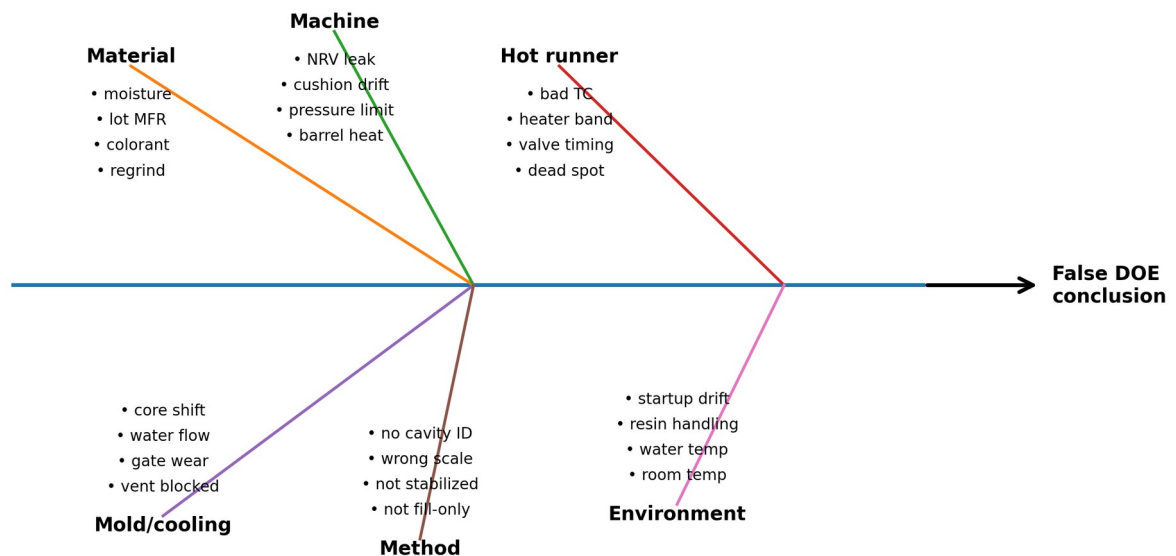


Figure 7 - Major confounders that can cause false DOE conclusions.

9.1 Step-by-step decision logic

Step	Question	Action if YES	Action if NO
1	Is the material condition controlled?	Proceed to machine stability checks.	Stop DOE. Correct material, drying, lot, color, and feed

2	Is the machine delivering repeatable shot volume?	Proceed to hot runner health check.	consistency. Run cushion/NRV study. Repair machine-side issue first.
3	Are all hot runner zones mapped, heating, and reading correctly?	Proceed to baseline flat-profile short shots.	Repair wiring, heaters, thermocouples, zone map, or controller settings.
4	Is the imbalance pattern stable by cavity or flow group?	Run temperature DOE or mechanical inspection based on pattern.	Investigate shot-to-shot variation, valve timing, material feed, and thermal stabilization.
5	Does the imbalance respond predictably to temperature?	Model and confirm smallest effective profile.	Look for gate/runner/vent/valve/cooling/tooling issues.
6	Does confirmation pass full-cycle quality?	Document profile and reaction plan.	Revisit pack/hold, cooling, valve timing, or tooling; do not release on short-shot data alone.

9.2 Common pitfalls and countermeasures

Pitfall	Why it creates a false result	Countermeasure
Running DOE before heat soak	Manifold temperature distribution changes during the trial.	Stabilize until PV/output/cycle/part weights are stable.
Using setpoints only	Bad thermocouple or swapped zones means setpoint is not actual temperature.	Record PV, output, alarm history, and independent checks where practical.
Changing fill speed mid-study	Changes shear heating and viscosity.	Lock fill profile. If speed is a factor, make it explicit.
Using full parts instead of short shots for fill balance	Pack/hold can hide fill imbalance.	Use fill-only short shots for balance, then full parts for validation.
No cavity IDs	Data cannot be mapped to physical flow paths.	Mold, label, or collect by cavity every shot.
Too few shots	Random noise looks like a factor effect.	Use repeat shots and calculate standard deviation/CV.
Ignoring valve gates	A delayed or sticking pin can dominate fill.	Verify valve operation and timing; run a valve timing DOE if needed.
Over-tuning temperature	Creates dissimilar melt conditions, degradation risk, or narrow process window.	Use minimal offsets and escalate large required offsets.
No confirmation after maintenance	Gate cleaning, heater replacement, or valve service changes the system.	Run reduced confirmation any time relevant hardware changes.

9.3 Environmental and operational criteria before a proper study

- Material lot, color, regrind, and drying parameters locked and recorded.
- Mold water temperature, flow, and hookup verified against the validated setup.
- Hot runner controller zone map verified and alarms clear.
- Machine nozzle contact force, nozzle tip, sprue bushing interface, and drool/stringing condition verified.
- Mold fully clamped and seated; no parting-line debris; vents cleaned.
- Robots, conveyors, grinders, and sprue pickers set so they do not vary cycle time or mix cavities.
- Scale calibrated and able to resolve meaningful cavity-weight differences.
- Operator, technician, and engineer agree on exact change plan and sample labeling.

10. References and Further Reading

Use these as starting points for validation, training, and deeper research. Supplier recommendations supersede generic guidance when processing a specific resin or hot runner system. Some public forum sources are included because they show practitioner reasoning, but they should not be treated as peer-reviewed technical authority.

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Appendix A - Hot Runner Balance/Temp DOE Data Sheet Template

Field	Required entry
Tool / mold ID	
Part number / revision	
Resin / grade / lot / color	
Dryer temp / dew point / dry time / moisture result	
Machine ID / barrel size / screw type	
Hot runner controller ID / program name	
Zone map verified by	
Mold water temp / flow / delta-T	
Fill time / injection speed profile	
Shot size / transfer / cushion	
Pack/hold disabled for fill-only?	
Scale ID / calibration status	
Cavity ID method	
Trial order and setpoints	
Actual PV/output records attached?	
Short-shot photos attached?	
Final process released by	

Appendix B - Example Cavity-Weight Calculation

Example: eight fill-only short-shot cavity weights are 8.31, 7.94, 8.07, 7.72, 8.54, 7.86, 8.23, and 7.58 g. Average = 8.031 g. Maximum = 8.54 g. Minimum = 7.58 g. Fill imbalance by the range/average method = $(8.54 - 7.58) / 8.031 \times 100 = 11.95\%$.

After a confirmed profile, the weights are 8.07, 8.11, 8.05, 8.01, 8.14, 8.09, 8.03, and 8.06 g. Average = 8.070 g. Maximum = 8.14 g. Minimum = 8.01 g. Fill imbalance = 1.61%. This is a meaningful improvement only if the full-cycle part data, visual quality, dimensions, resin stability, and process window are also acceptable.

Appendix C - Red Flags Requiring Tooling/OEM Escalation

- One cavity requires a much higher tip temperature than adjacent cavities to fill properly.
- Controller output is saturated or erratic on a zone.
- Zone PV does not respond logically to setpoint changes.
- Balance changes after short stops or startup but recovers later.
- Hot runner leaks, drools, strings, or freezes at a tip.
- Valve gate stroke or timing differs cavity-to-cavity.
- Gate steel shows wear, damage, flash, obstruction, or witness marks.
- Short-shot pattern is inconsistent shot-to-shot with stable setpoints.
- Balanced fill creates unacceptable dimensions, warp, weld-line weakness, gloss, or degradation.
- Required offsets exceed the resin supplier or hot-runner OEM processing guidance.

Appendix D - Detailed Shop-Floor SOP for Running the Study

This SOP is written for production execution. It assumes the engineering team has already selected the DOE factors and allowable low/nominal/high settings. The processor or technician executing the study should not be expected to invent the experimental design at the press. The job at the press is disciplined execution, traceability, and containment.

Phase	Owner	Minimum tasks	Hold point
Pre-shift planning	Process engineer / tooling engineer	Define objective, cavity numbering, factors, levels, run order, sample size, scrap containment, quality checks, maximum allowable temperature offsets, and stop criteria.	Do not start if factor levels exceed resin or hot-runner OEM limits.
Tool installation	Setup technician	Install mold, verify clamp, water, air, hydraulics, valve-gate connectors, electrical connectors, robot, safety interlocks, and hot-runner controller map.	Do not heat hot runner if zone map is unknown or unlabeled.
Material preparation	Material handler / technician	Confirm resin, lot, colorant, regrind, drying temperature, dew point, hopper level, and purge material compatibility.	Do not run if material identity or drying state is unknown.
Thermal stabilization	Processor	Bring barrel, nozzle, hot runner, and mold water to baseline. Confirm all PVs, outputs, water flow, and alarms. Run stabilization cycles.	Do not collect data during heat-soak drift.
Baseline short-shot	Processor / quality	Disable pack/hold for fill-only baseline, set repeatable shot size, collect cavity-identifiable short shots, weigh and photograph.	Stop if cavity IDs are mixed or parts break/trap in tool.
DOE trials	Processor / engineer	Execute trials in approved order. Change only listed factors. Wait for thermal stabilization after each change. Collect repeat shots.	Stop if a zone alarms, material degrades, or pressure approaches unsafe limit.
Analysis at press	Engineer / technician	Calculate imbalance and check response direction. Verify that changes make physical sense by flow group.	Do not chase a single noisy shot.
Confirmation	Processor / quality	Run predicted profile in fill-only and full production cycle. Confirm weights, dimensions, visuals, cavity pressure if available.	Do not release on fill-only data alone.
Release / containment	Quality / production	Revise setup sheet, define reaction plan, release accepted product, quarantine trial product per quality rules.	No undocumented profile changes after release.

Appendix E - Measurement System Analysis for Cavity Balance

Cavity balance is a measurement problem before it is a process problem. The study is only as good as the scale, cavity identification, sampling method, and repeatability of the injection event. A scale that resolves 0.1 g may be adequate for a 40 g automotive part and useless for a 0.35 g medical micro part. Define measurement resolution before the DOE.

Measurement item	Minimum practice	Engineering note
Scale resolution	Use a scale with resolution at least 10 times finer than the expected meaningful cavity difference. For small parts, use analytical or semi-micro scales with draft shielding.	If the target is to detect 0.005 g differences, a 0.01 g scale is not acceptable.
Scale calibration	Verify calibration status and use check weights before and after the study.	A drifting scale can create artificial thermal response.
Sample temperature	Cool samples consistently before weighing or weigh immediately with consistent time delay.	Hot short shots can continue shrinking or losing volatiles; timing matters for small parts.
Cavity identification	Use molded cavity IDs, marked trays, cavity-specific bags, or robot drop-map confirmation.	Do not sort by visual location after parts are mixed.
Sample count	Use a minimum of five stable shots per	A single shot cannot define a cavity response.

	condition for screening; use more when CV is high or part weight is low.	
Outlier handling	Document outliers with physical cause only: stuck part, damaged sample, mixed cavity, obvious purge contamination.	Do not delete data just because it weakens the conclusion.
Response rounding	Keep raw data at full scale resolution; round only in reports.	Rounding early can hide small but relevant trends.

Recommended calculations include average by cavity, standard deviation by cavity, coefficient of variation by cavity, overall range/average imbalance, maximum deviation from average, shot-to-shot total shot weight variation, and rank stability. Rank stability means the same cavities remain high or low across repeated shots and trials. If the rank order constantly changes, the process is not stable enough for a temperature DOE.

Appendix F - Statistical DOE Planning Details

A hot runner temperature DOE should not be overcomplicated, but it must be statistically defensible. The simplest valid plan is often a structured screening design with center points and confirmation. Full factorial designs become impractical with many zones; eight nozzle zones at three levels would require 6,561 conditions before replications. That is not a production-friendly plan. Group zones by physics, not convenience.

DOE design	Number of factors handled well	Strength	Weakness	Recommended use
One factor at a time	1-3	Easy to execute and explain.	Cannot detect interactions and is vulnerable to time drift.	Initial diagnosis on low-risk tools.
2-level factorial	2-5	Estimates main effects and selected interactions.	No curvature unless center points are added.	Engineering screening.
Fractional factorial	4-8+	Efficient for high cavitation or many zone groups.	Effects can be aliased; needs knowledgeable design.	High-cavitation tools with limited press time.
Response surface / central composite	2-4	Models curvature and optimum region.	More trials and more stabilization time.	Critical parts or nonlinear thermal response.
Taguchi / robust parameter design	3-8	Good for screening control/noise factors.	Can be misused if interactions are ignored.	When noise factors such as lot or water temp can be deliberately blocked.
Sequential Bayesian / adaptive DOE	2-8	Efficient when software support exists and each trial is expensive.	Requires validated algorithm and guardrails.	Advanced engineering teams with reliable data infrastructure.

Design rule	Practical interpretation
Block by time	If the study takes several hours, separate results by time block to avoid interpreting heat soak or material drift as a factor effect.
Use center points	A nominal center point repeated early, middle, and late shows whether the baseline moved.
Randomize carefully	Randomization is statistically preferred, but large temperature moves can take too long to stabilize. Use restricted randomization or blocking when required.
Define safe levels	Low and high levels should be large enough to create measurable response but small enough to protect resin, hot runner, and part quality.
Use actual PV data	Analyze actual temperature when possible; the process responds to delivered temperature, not the planned setpoint.
Preserve raw data	Keep the full trial order, raw weights, and actual process readings. Summary charts alone are not enough for future troubleshooting.

Appendix G - Component-Level Hot Runner Diagnostic Map

Component	What can go wrong	Symptom during balance DOE	Verification method	Typical corrective action
Manifold channel	Machining mismatch, plug misalignment, dead spot, internal obstruction, leakage, poor insulation.	Cavities grouped by branch run consistently high/low or drift after heat soak.	Thermal map, pressure-loss comparison, OEM review, teardown inspection if justified.	Clean/repair manifold, correct plug, redesign flow path, improve insulation.

Nozzle/drop	Heater failure, thermocouple error, tip damage, poor contact, cold slug, drool.	One cavity very sensitive to tip temperature or unstable at startup.	Zone PV/output trend, resistance check, thermocouple check, gate/tip visual inspection.	Replace heater/TC/tip, clean drop, repair gate insert.
Gate insert	Wear, erosion, burr, contamination, mismatch to cavity.	One cavity flashes or fills early; another shorts. Gate vestige differs.	Microscope, pin gauges, replica, visual comparison to baseline.	Repair/replace insert, polish, restore dimensions.
Valve pin	Sticking, delayed open/close, bent pin, worn seal, hydraulic/pneumatic/electric response variation.	Cavity order changes, intermittent short shot, gate vestige and pressure trace inconsistent.	Stroke test, timing trace, pressure overlay, actuator diagnostics.	Service actuator/pin, correct timing, replace seals.
Thermocouple	Swapped, loose, poorly located, shorted, open, reading steel not melt-relevant temperature.	Temperature change appears ineffective or opposite of expected.	Zone map test, ice/heat check when safe, resistance, controller diagnostics.	Repair wiring, relocate/replace TC, correct zone map.
Heater band/cartridge	Open circuit, partial output, poor contact, uneven heating.	Zone output high, slow recovery, frequent alarms, local short shots.	Resistance, amp draw, controller output, thermal imaging.	Replace heater, improve contact, correct wiring.
Controller	Wrong program, slave mode, PID poorly tuned, alarm disabled, connector issue.	Actual PV unstable or not matching setpoint; profile not repeatable across startups.	Audit program, autotune status, alarm bands, output trends.	Correct program, tune zones, repair connectors, restore alarm policy.
Wiring/connectors	Crossed zones, intermittent open, damaged pins, contamination.	DOE response assigned to wrong cavity or intermittent jumps.	Continuity check, wiggle test under safe conditions, zone map verification.	Repair cables/connectors, label harnesses.

Appendix H - Material-Specific Deep Dive

The material section below is intentionally practical. It does not replace a resin datasheet. Its purpose is to help processors predict how a hot runner temperature DOE can mislead them when a polymer family has a distinctive rheological or thermal behavior.

Material family	Hot-runner DOE sensitivity	Defects to watch	What not to do
Polypropylene (PP)	Often responds strongly to shear and temperature but tolerates moderate thermal moves. Semi-crystalline shrink and warpage can change even when fill looks balanced.	Warp, sink, dimensional spread, gate blush, flow marks, flash on easy-flow cavities.	Do not release based only on short-shot balance; confirm shrink/warp and sealing function.
Polyethylene (HDPE/LDPE/LLDPE)	Broad window but grade-to-grade MFR differences are large. Hot runner hold-up and colorant dispersion can affect appearance.	Splay from contamination, color streaks, gate stringing, sink, flash.	Do not assume a PE profile transfers across MFR grades or color packages.
ABS	Amorphous with cosmetic sensitivity. Higher temperature may improve fill but change gloss and increase volatiles or splay.	Gloss mismatch, blush, weld lines, splay, burning, stress whitening.	Do not use tip heat to hide poor venting or gate restrictions on cosmetic parts.
Polycarbonate (PC)	High viscosity; temperature has strong effect. Drying and residence time are critical.	Splay, bubbles, yellowing, brittleness, gate blush, high stress.	Do not exceed hot runner limits to force balance; investigate pressure loss and gate sizing.
PC/ABS	Blend behavior depends on grade and additive package. Thermal history affects appearance and impact.	Splay, silver streaks, gloss mismatch, weld weakness, color shift.	Do not treat as either pure PC or pure ABS; use supplier-specific window.
Nylon (PA6/PA66)	Moisture changes viscosity and properties. Semi-crystalline freezing can make gate/tip design important.	Splay, voids, dimensional moisture effects, burn, brittle degraded material.	Do not run a long high-temperature study without residence-time control.
PBT/PET	Semi-crystalline polyesters; moisture and temperature control affect hydrolysis, crystallization, and dimensions.	Splay, brittleness, flash, dimensional drift, gate blush.	Do not run wet material and interpret lower viscosity as a good balance result.
POM/acetal	Thermally sensitive; degradation products are hazardous. Hot spots and long dwell are serious risks.	Odor, gas, burn, plate-out, black specks, brittleness.	Do not experiment outside supplier limits. Purge and ventilate per material guidance.
PVC	Very heat/shear sensitive. Fill optimization must avoid degradation.	Discoloration, gas, burn, corrosion, plate-out.	Do not use broad high-temperature offsets as a balance strategy.
LCP	Highly orientation-sensitive and fast filling. Small geometry differences can drive large property variation.	Weld weakness, anisotropic dimensions, jetting, burn.	Do not rely on weight balance alone; evaluate orientation-dependent performance.
TPE/TPU	Can be shear and temperature sensitive; drool/stringing can be dominant.	Stringing, sticking, surface drag, gate marks, degradation.	Do not ignore demolding and gate shutoff while focusing on fill weight.

Appendix I - Case Study Narratives for Training

The following training narratives are built from documented mechanisms and public cases. They are written as composite teaching cases, not as private customer reports. Use them to train technicians on decision logic.

I.1 High-cavitation closure mold: stable cavity-to-cavity weight spread

A 32-cavity cap mold shows four cavities consistently 5-7% light during fill-only short shots. Full-cycle parts pass weight because hold pressure is high, but dimensional capability on the sealing feature is poor in the same cavities. A flat hot-runner profile baseline is run after confirming melt temperature, cushion, and water flow. The light cavities share one branch of the manifold. Increasing the branch manifold zone by 10 F improves the light group but also increases gate blush and slightly shifts the weld line. A tip-zone-only change has little effect. Tool inspection finds a partially obstructed branch plug and insulation damage. After repair, the flat profile balance improves enough that only a 3 F branch offset is needed.

Lesson: the DOE did its job because it revealed a branch-level pattern and prevented permanent over-tuning. The win was not the temperature change; the win was finding the mechanical defect.

I.2 Medical component: short-shot balance passes, full-cycle validation fails

A four-cavity medical part is balanced from 8% to 2% short-shot imbalance by raising two tip zones. Initial enthusiasm is high. Full-cycle validation then shows different dimensions and different gate blush by cavity. Cavity pressure traces reveal that two cavities fill simultaneously but pack differently because their gates remain open longer due to the higher local thermal condition. Gate-seal study is repeated and the hold profile is revised. The final released process uses a smaller tip offset, a corrected hold-pressure profile, and tighter gate-seal monitoring.

Lesson: fill balance is not final process validation. Hot runner temperature changes also affect gate seal and packing.

I.3 Automotive cosmetic part: balance problem is actually venting

A two-drop hot-runner tool produces a visible flow mark and occasional burn in the same region. The initial assumption is that one tip is cold. A short-shot progression shows the affected area is end-of-fill and traps air. Raising the tip temperature reduces short shots but worsens burn marks. Cleaning and enlarging vents corrects the filling pattern without a hot-runner profile change.

Lesson: trapped air can look like low melt temperature. Short-shot photos and vent inspection should precede thermal tuning.

I.4 Consumer-goods family tool: thermal balance is the wrong control mode

A family tool with unequal part volumes cannot be balanced with normal temperature profiling because the cavities fundamentally require different flow volumes and pressure histories. Large thermal offsets create part-to-part material-property differences. The tool is converted to valve-gate sequence control and cavity-pressure monitoring. Balance improves without forcing every cavity to see the same thermal compensation.

Lesson: family tools often need flow or pressure control, not simply hot-runner temperature offsets. Published valve-gate sequencing work makes the same point: thermal balancing can improve fill timing while creating dissimilar melt temperatures [R5].

Appendix J - Process Parameter Interaction Matrix

Changed parameter	Observed effect on balance	Likely mechanism	Action before accepting conclusion
Increase fill speed	Some cavities fill faster; imbalance may increase or decrease depending on shear distribution.	Higher shear rate lowers viscosity and creates viscous heating; shear-heated layers split unevenly.	Compare to viscosity curve and pressure limit. Keep speed fixed during temp DOE.
Increase melt temperature	All cavities may fill heavier; cold/restricted cavities may improve more.	Lower viscosity and delayed freeze.	Verify actual melt, residence time, degradation, and dimensions.
Increase mold temperature	Less early freeze, improved surface, changed gate seal.	Lower frozen-layer growth and altered heat transfer.	Document water flow and run full-cycle confirmation.
Raise one tip temperature	Local cavity fills faster/heavier; gate seal may extend.	Lower local pressure loss and higher local melt temperature.	Check gate appearance and gate-seal timing.

Hot Runner Balance / Temperature DOE Technical Reference - Technical draft, validate before production release

Lower one tip temperature	Local cavity fills slower/lighter; may reduce drool or flash.	Higher local viscosity and earlier freeze.	Do not create cold-slug or gate-freeze defects.
Increase back pressure	Melt homogeneity may improve, melt temperature may rise.	More work on melt during recovery.	Reverify melt temperature and recovery time.
Increase screw RPM	Recovery faster, melt shear heating may increase or melting may worsen depending screw/resin.	Mechanical shear and residence profile changed.	Do not alter during DOE unless it is a factor.
Increase pack pressure	Part weights balance may appear better, but fill imbalance can remain.	Late cavities get forced after fill if gates remain open.	Use fill-only balance and cavity pressure to separate fill from pack.
Increase clamp force	Flash may reduce but venting may worsen if vents close.	Parting-line condition changes.	Do not use clamp to mask fill or venting problems.

Appendix K - Hot Runner Temperature Profile Release Criteria

A profile should not be released simply because the newest chart looks better. Release requires evidence that the process is robust, safe, repeatable, and maintainable by the production team.

Release criterion	Minimum expectation	Evidence
Balance improvement	Meets project-specific imbalance target or demonstrates statistically significant improvement without new defects.	Short-shot weight table, before/after chart, raw data.
Repeatability	Cavity rank and weights stable across repeated confirmation shots.	Mean, standard deviation, CV by cavity.
Full-cycle quality	Weights, dimensions, visual defects, functional tests, and capability meet requirements.	Quality report by cavity.
Thermal safety	All setpoints and actual PVs within resin and hot-runner OEM limits.	Process sheet and controller trend.
Material integrity	No degradation indicators: splay, odor, color shift, black specks, plate-out, brittleness.	Visuals, material tests if critical.
Gate seal/pack	No cavity-specific overpack/underpack introduced by profile.	Gate-seal study or cavity pressure overlay.
Startup repeatability	Profile repeats after cold start or defined startup procedure.	Startup confirmation record.
Maintenance plan	Zone map, spare heaters/TCs, valve service notes, cleaning plan documented.	Tooling/maintenance signoff.
Reaction plan	Operators know what to check before changing temperatures.	Controlled setup sheet and escalation matrix.

Appendix L - Training Notes for New Processors

The next generation of molders must learn that hot runner balancing is not magic and not guesswork. It is controlled measurement. The mold tells the truth when the process is stable, the samples are labeled, and the data is recorded honestly. The mold lies when the material is wet, the cushion is drifting, the vents are dirty, the hot runner has alarms, and the technician changes three things at once.

Principle	Plain-language explanation	Technical reason
Short shots first	Look at how the mold fills before packing hides the evidence.	Pack pressure can force late cavities full and hide fill imbalance.
Flat profile baseline	Start with all comparable zones at nominal before profiling.	Without a baseline, you do not know what the tool naturally does.
Cavity ID is mandatory	A weight without cavity number is nearly useless.	Balance is spatial; data must map to flow path.
Small offsets are better	A little temperature trim can be normal; a big trim is a clue.	Large thermal offsets create different melt histories and can mask steel defects.
Actual temperature matters	Setpoint is a request; PV/output tells what the system did.	Bad sensors or heaters can make setpoints misleading.
Do not chase noise	One odd shot is not a pattern.	DOE decisions require replication and repeatability.
Tooling beats knob-turning	If steel is wrong, the process will fight you forever.	Robust process windows come from correct geometry, venting, cooling, and hardware.

Appendix M - Additional References Addendum

Ref	Source
R25	Autodesk Support, "How to set up and run Runner Balance analysis in Moldflow" - describes runner balance analysis as a tool for optimizing runner section volume for balanced filling.
R26	Mold-Masters hot-runner user and troubleshooting manuals - use for system-specific safety, wiring, heater, thermocouple, valve, and maintenance requirements.
R27	Resin supplier technical datasheets and processing guides - final authority for melt temperature, mold temperature, drying, residence time, purging, and degradation limits.
R28	Internal quality system procedures - final authority for trial part segregation, process change approval, capability requirements, and medical/regulated product documentation.